

Aditya Reddy Bondugula

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SUMMARY

- Results-driven Data Engineer with 7+ years of experience designing, building, and optimizing large-scale data pipelines across cloud, big data, and enterprise environments
- Skilled in architecting ETL/ELT workflows using Spark, PySpark, Hive, SQL, AWS, Azure, and Hadoop to support analytical, regulatory, and operational reporting
- Experienced in migrating legacy pipelines to cloud-native architectures leveraging AWS EMR, Glue, Lambda, S3, Redshift, and Azure Data Factory
- Strong background in building real-time streaming solutions using Kafka, Kinesis, and Spark Streaming for fraud detection, risk scoring, and operational intelligence
- Adept at data modeling, data lake design, and implementing scalable batch/streaming solutions for multi-terabyte datasets. Proven track record in performance tuning, cluster optimization, and cost-efficient compute scaling across EMR and Databricks
- Skilled in developing automated data quality frameworks, ensuring accuracy, integrity, and governance across enterprise pipelines
- Collaborated closely with cross-functional teams to translate business rules into robust technical solutions. Experienced in CI/CD automation using Git, Terraform, Jenkins, and DevOps practices for reliable production deployments
- I am passionate about building clean, secure, high-impact data systems that enable decision-making, analytics, and machine learning

WORK EXPERIENCE

Data Engineer | AAA Texas | Texas, USA

Mar 2025 – Present

- Engineered distributed data processing pipelines on AWS EMR and Databricks to handle multi-terabyte customer and transaction datasets
- Developed PySpark and Scala pipelines to process agent performance, member transactions, and policy servicing data for operational analytics
- Designed low-latency streaming pipelines for member activity logs, claim updates, roadside assistance requests, and agent workflow events
- Built a scalable data lake structure with raw, refined, and curated zones, implementing partitioning and lifecycle policies for cost-efficient storage
- Implemented robust data validation suites using tools like Great Expectations/PyDeequ to ensure consistency and reliability of high-volume datasets
- Created optimized Spark jobs using intelligent caching, broadcast joins, and memory tuning to handle compute-heavy financial transformations
- Implemented Delta Lake features such as ACID transactions and Z-ordering to accelerate query performance for analytics and machine-learning workloads
- Developed internal APIs and high-throughput data connectors using Python and Node.js to integrate partner feeds, bureau data, and microservices
- Set up secure identity and access flows using OAuth2, JWT, and Keycloak to control and audit access to sensitive banking datasets
- Wrote SQL logic in Redshift to support agent scorecards, member retention dashboards, and multi-branch operational analysis.
- Created reusable Terraform blueprints for provisioning EMR clusters, network components, IAM roles, and all core data engineering infrastructure

- Leveraged Argo Workflows and Temporal to orchestrate multi-step pipelines with complex dependencies, retries, and distributed job checkpoints
- Designed pipeline observability dashboards using CloudWatch logs, metrics, and alarms to proactively detect abnormalities in data flows
- Applied Spark performance best practices including auto-scaling, optimized shuffle configurations, and partition pruning across heavy workloads
- Built QuickSight dashboards visualizing member engagement metrics, agent productivity KPIs, claims turnaround performance, and regional operations insights
- Developed secure connectors to integrate internal AAA policy systems, membership databases, claims platforms, and third-party partner data sources
- Led internal GenAI initiatives by integrating Amazon Bedrock/OpenAI models to automatically generate data quality summaries, lineage metadata, and pipeline documentation

Data Analytics Engineer | BJC HealthCare | Missouri, USA

Feb 2024 – Dec 2024

- Designed and configured end-to-end Azure Data Factory (ADF) pipelines, including linked services, parameterized datasets, and incremental data flows for multi-source integrations
- Developed optimized PySpark transformation scripts in Databricks to support large-scale data preparation, enrichment, and table-to-table operations across analytical layers
- Built scheduled and near-real-time ingestion jobs to integrate donor contributions, fundraising event logs, and CRM updates
- Optimized Databricks workflows by analyzing job performance metrics and recommending cost-efficient compute configurations
- Developed transformation logic using PySpark and Data Factory mapping flows for donor, fundraising, and patient-support datasets
- Integrated ADF pipelines with diverse source systems such as SFTP, Informix, Sybase, and cloud storage, ensuring reliable ingestion and schema consistency across environments
- Led troubleshooting efforts for pipeline failures, conducted data validation, and documented engineering standards and workflow designs to improve delivery efficiency

BigData Engineer | Deloitte India Offices of US | Telangana, India

Jun 2020 – Oct 2022

Sub Project 1

- Led the design of distributed data processing platforms using Spark, EMR, Databricks, Hive, and Airflow to power enterprise-scale HR analytics across Deloitte's global workforce
- Architected real-time event-driven ingestion pipelines via Kafka & Spark Streaming for employee lifecycles, onboarding, attrition, and staffing signals consumed by advanced ML models
- Tuned Spark clusters and workloads (YARN, EMR, Databricks) by optimizing partition strategy, DAG design, GC tuning, and cluster autoscaling reducing compute cost by 30%
- Engineered complex ETL/ELT transformations combining PySpark, Scala, SQL, and AWS Glue for compensation, engagement, and leadership dashboards
- Migrated multiple legacy pipelines from Informatica and traditional SQL workloads into Spark-based orchestration, improving pipeline stability and latency
- Implemented audit logging, data lineage, and access governance frameworks using AWS Lake Formation and Atlas, ensuring secure and compliant data movement
- Led code reviews, mentored junior data engineers, established engineering standards, and developed reusable modular components improving team productivity by 50%

- Built CI/CD automation for Spark and Glue jobs using Jenkins/GitLab/Terraform to standardize production deployments across environments
- Designed scalable ingestion paths for third-party HR tools (Workday, SuccessFactors, Greenhouse) combining API, Kafka, and batch connectors for unified workforce data
- Collaborated with the ML & People Insights teams to produce feature-ready datasets for predictive attrition models, leadership scoring engines, and workforce forecasting

Sub Project 2

- Built large-scale data processing pipelines on Hadoop and EMR using Spark (PySpark/Scala), Hive, and Kafka to support enterprise workforce analytics and operational dashboards
- Converted legacy SQL and Hive scripts into optimized Spark transformations using DataFrames, RDDs, and Spark SQL, reducing compute time by 40%
- Designed and developed Spark Streaming pipelines to ingest HR, staffing, and project activity data in real time for internal Deloitte analytics platforms
- Implemented data lake architecture using AWS S3 and curated bronze-silver-gold layers for People Analytics data products
- Performed performance tuning across Spark jobs optimizing shuffles, caching strategies, cluster resources, and executor configurations
- Conducted advanced data modeling for attrition, workforce planning, compensation, and promotion datasets, enabling ML-ready standardized datasets
- Partnered with cross-functional HR and business stakeholders to translate workforce insights into scalable pipelines aligned to internal governance and compliance
- Developed automated quality validation frameworks using PyDeequ and custom Spark checks to ensure accuracy of HR and workforce metrics before downstream consumption
- Integrated People Analytics pipelines with Tableau/Power BI datasets, enabling near real-time visualization of staffing KPIs and organizational health indicators
- Implemented job orchestration and modular workflow dependencies using Airflow and custom Spark operators, improving reliability of daily HR data refresh cycles

CHARTER BANK | Telangana, India

May 2016 – Jun 2020

Sub Project 1 - ETL Developer

- Designed and maintained end-to-end ETL workflows to integrate data from Oracle, SQL Server, and flat files into Charter Bank's centralized data warehouse, ensuring high-quality and timely data availability
- Designed and maintained complex ETL workflows using AWS Glue, Lambda, and Step Functions to move data across operational and analytical systems
- Developed PySpark transformation scripts to standardize account, loan, and transaction records for downstream financial reporting
- Built automated data ingestion pipelines from APIs, S3 buckets, and on-prem SQL servers to ensure timely availability of operational data
- Implemented robust validation checks in Python to monitor schema drift, data volume anomalies, and field-level inconsistencies
- Created reusable ETL templates for incremental loads, CDC patterns, and SCD Type-2 updates across multiple datasets
- Migrated legacy Informatica mappings into modern Glue and PySpark ETL jobs, ensuring feature parity and improved performance

- Developed parameterized ETL workflows to support multi-environment deployments across dev, QA, and production
- Worked closely with business teams to translate reporting requirements into ETL logic for dashboards and audit extracts
- Built data cleansing routines to handle null handling, type conversions, address standardization, and customer record matching
- Automated ETL run monitoring with CloudWatch metrics and custom SNS alerts for job failures and SLA breaches
- Optimized Glue job execution by tuning worker types, dynamic frames, and partitioning strategy to reduce overall compute cost
- Created lineage documentation and ETL flow diagrams to help auditors and engineers understand data movement
- Performed root-cause analysis for broken ETL jobs, data gaps, or duplicate records across staging and curated layers
- Managed version control for transformation logic using Git branching standards and CI/CD pipelines
- Integrated ETL outputs with Redshift, Snowflake, and S3 data zones for analytics, dashboarding, and regulatory teams
- Coordinated cross-team data validations with Finance, Risk, and Compliance teams to ensure accuracy before publishing curated datasets

Sub Project 2 - Data Engineer

- Designed and maintained end-to-end Azure data pipelines leveraging HDInsight, Databricks, Data Lake Storage, Blob Storage, Data Factory, Synapse, Azure SQL, and SQL Data Warehouse
- Built a scalable enterprise data lake using ADF and Blob Storage, enabling analytics, machine-learning workflows, and complex data processing across business units
- Integrated structured and semi-structured data from MongoDB, MS SQL Server, and Azure SQL using ADF connectors, SQL APIs, and Mongo APIs for unified data availability
- Developed PySpark-based transformation jobs for large-scale data mining, cleansing, and real-time analytical insights
- Improved data quality and processing performance using advanced Python techniques, including higher-order functions, lambda logic, and pattern-matching transformations
- Applied Databricks and Spark best practices to clean, aggregate, and transform datasets for downstream analytics and BI reporting
- Used Azure Synapse to orchestrate processing workloads and serve optimized datasets for business intelligence and predictive modeling needs
- Created custom ingestion adapters using Spark, Sqoop, and Airflow to move high-volume data from traditional RDBMS systems into Azure Data Lake
- Enhanced query performance by redesigning data models and integrating Redis caching to accelerate Snowflake workload execution
- Developed automated, parameterized workflows for incremental data ingestion from relational databases into the cloud data lake
- Monitored and tuned Spark clusters using Azure Log Analytics and Ambari, improving performance and transitioning logging from SQL Server to CosmosDB
- Built Talend-based ETL jobs to load curated datasets into Snowflake, supporting reporting, dashboards, and executive analytics
- Managed resource scaling and job scheduling using Azure Kubernetes Service (AKS), ensuring efficient processing for compute-heavy workflows

- Implemented CI/CD automation, debugging practices, and monitoring dashboards using Azure DevOps to streamline releases and operational visibility
- Enforced enterprise-grade security and governance by configuring Azure Active Directory roles and Apache Ranger policies
- Collaborated with data scientists on feature engineering and operationalizing machine-learning models in production environments
- Tuned Spark NLP workloads by optimizing memory settings, batch intervals, and execution parallelism for faster text-processing pipelines
- Delivered reliable, cleaned, and analytics-ready datasets to Power BI teams, supporting dashboards for operations, finance, and business leadership

SKILLS

Data Engineering & Pipeline Tools: Apache Spark (PySpark, Scala, Spark SQL), Hadoop (HDFS, Hive, MapReduce, YARN), Kafka, Kinesis, Spark Streaming, Databricks, AWS EMR, Cloudera, Airflow, Oozie, NiFi, Talend, Informatica, DataStage, CI/CD: Jenkins, GitHub Actions, Azure DevOps, IaC: Terraform, CloudFormation

Cloud & Database: AWS: S3, EC2, EMR, Glue, Lambda, Redshift, Athena, Kinesis, IAM, CloudWatch, MySQL, SQL Server, Hubble, SharePoint, Teradata, Hadoop Hive, SAP Business Object, Azure: Data Factory, Databricks, Synapse, Data Lake, Blob Storage, GCP: BigQuery, Cloud Storage, Snowflake, Oracle, PostgreSQL, MongoDB, Cassandra, DynamoDB, Data modeling (Star/Snowflake schema), Fact/Dim design, OLAP/OLTP, Data lake architecture (Bronze/Silver/Gold)

Business Intelligence: Power BI, Tableau, AWS QuickSight, SSRS, Dashboard modeling, KPI frameworks, drill-downs, Data storytelling & executive reporting, Power Query, DAX, M-Formula, Smartsheet, Building BI semantic models & optimized datasets, Performance tuning for BI queries (DAX/SQL)

Data Analysis: Monarch, JIRA, Microsoft Access, Macros, Visio, Confluence, Business Analysis, Python (Pandas, NumPy), R, Advanced SQL (window functions, CTEs, optimization), Exploratory data analysis (EDA), Data profiling, data validation, data quality checks, Statistical modeling & data mining, Working with structured & semi-structured data (JSON, XML, Parquet, ORC), Feature engineering & data preparation for ML, root-cause analysis, Excel (Pivot Tables, Power Pivot, Index/Match, XLOOKUP)

EDUCATION

Webster University | Missouri, USA

Jan 2023 – Dec 2024

Master of Science, Data Analytics

CMR College of Engineering & Technology | Telangana, India

Jun 2012 – May 2016

Bachelor of Technology, Computer Science Engineering